

Performing a Basic Audit of Your AWS Environment – hands-on lab

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Lab overview

Task 1: Audit user permissions in IAM

AWS Identity and Access Management (IAM) allows you to securely control access to AWS services and resources for users (AWS, n.d). Using IAM enables you to create and manage AWS users and groups and user permissions to allow and deny their access to AWS resources.

In this lab, I launched IAM Secure AWS Access Control to review permissions, group assignments, and roles associated with the auditing instance.

- Step 1: In the search bar, search for and choose “IAM”
- Step 2: In the navigation pane at the left, under “Access management” choose “Users”
- Step 3: On the “Users” page, choose the link for “user-1” to view its details
- Step 4: Review the “Summary”
- Step 5: Select the “Security credentials” tab to review it

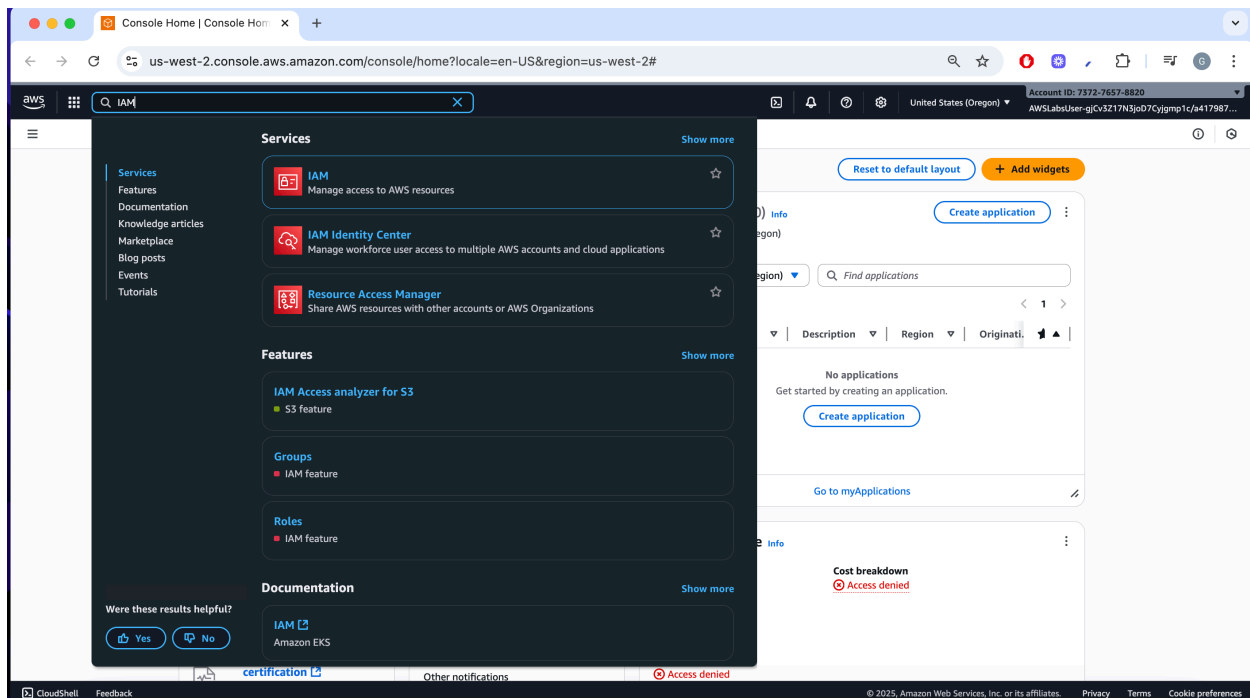


Figure 1: IAM search (source: personal collection)

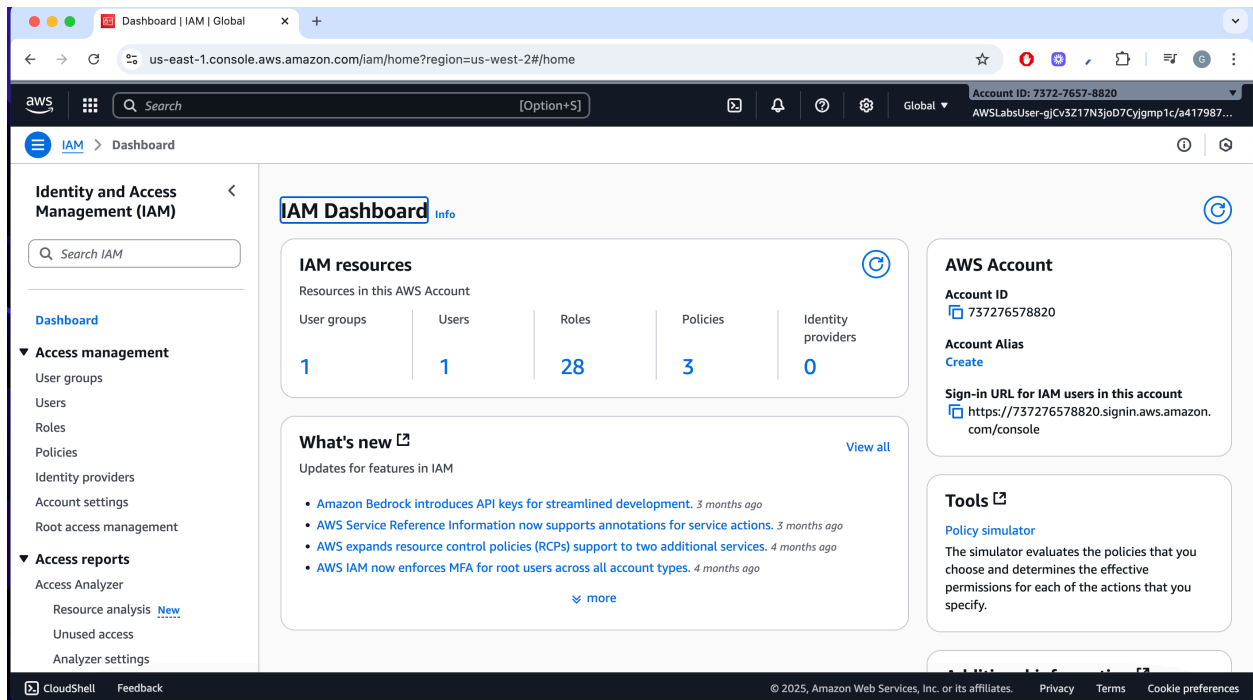


Figure 2: IAM Dashboard (source: personal collection)

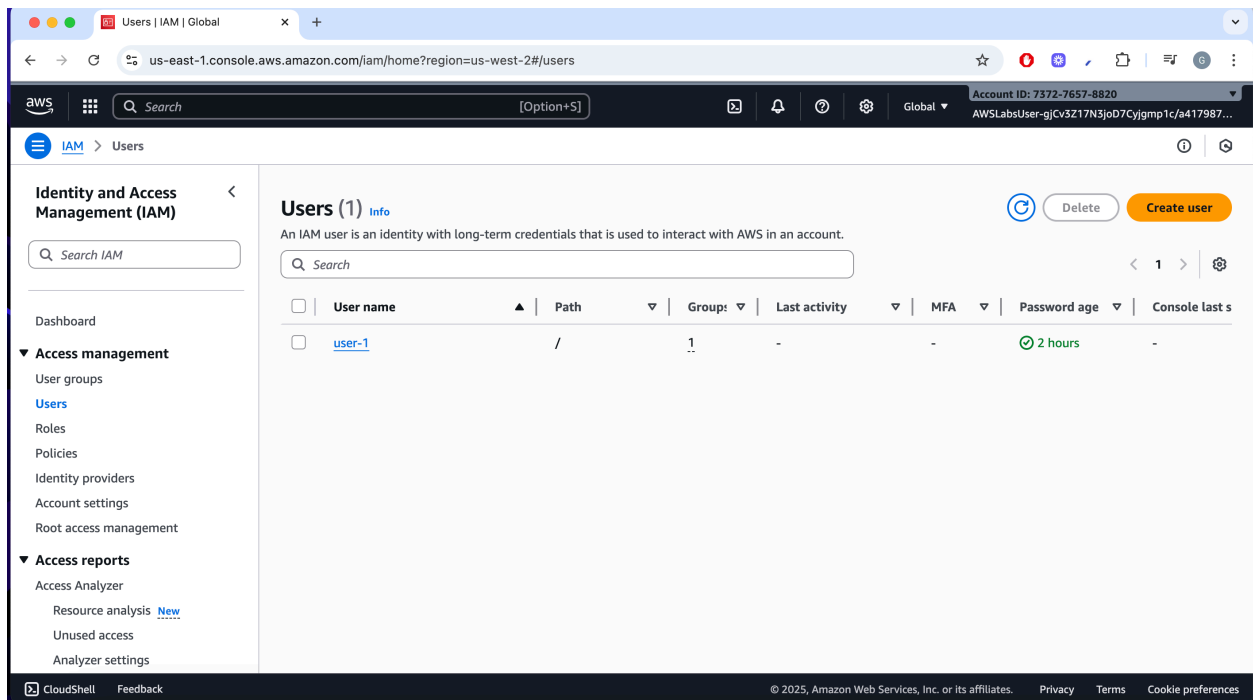


Figure 3: Users page dashboard (source: personal collection)

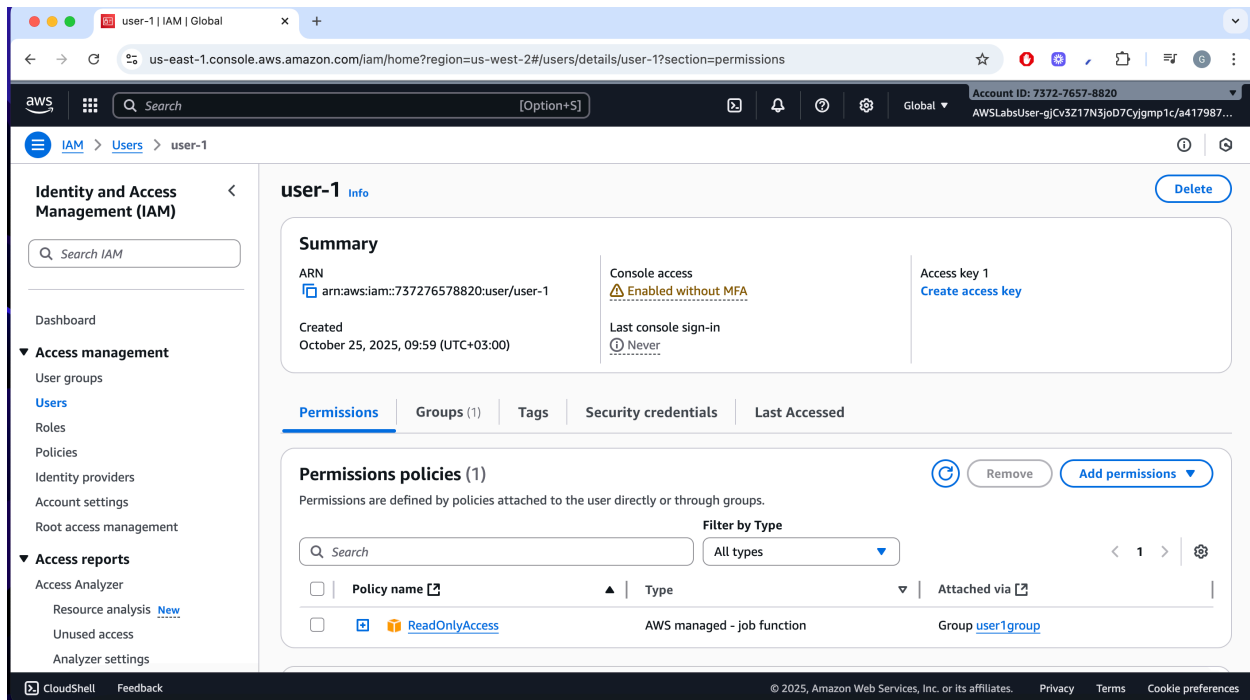


Figure 4: User-1 dashboard (source: personal collection)

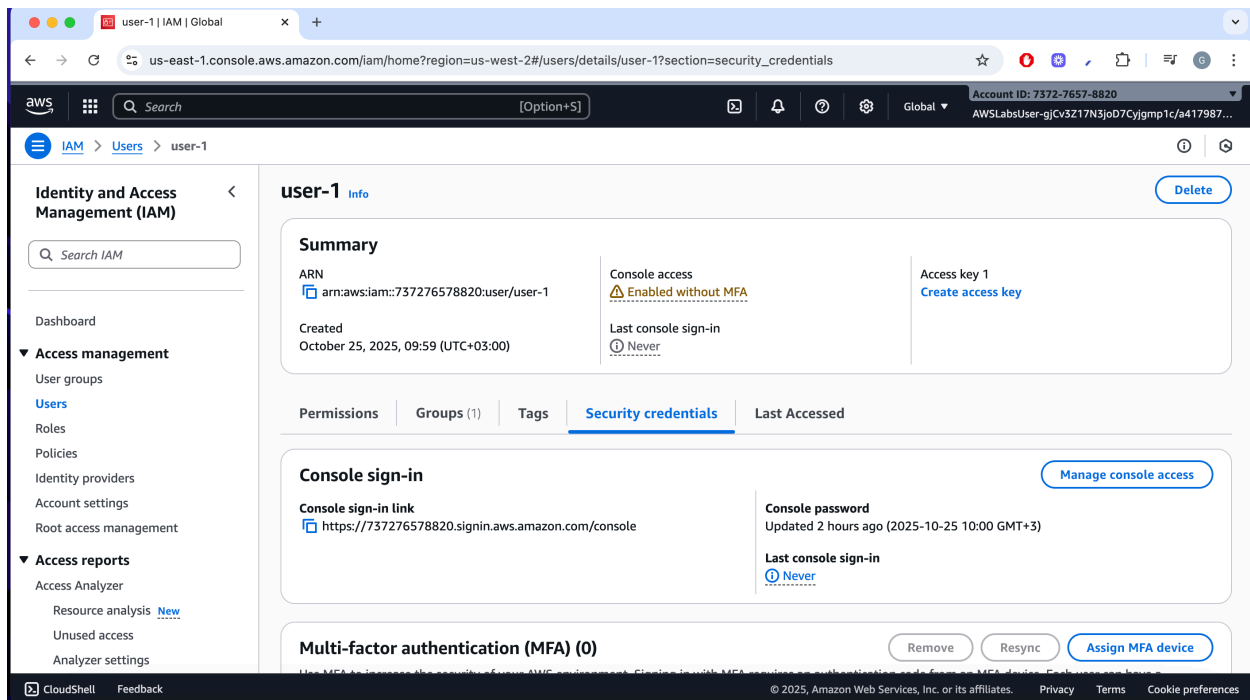


Figure 5: Security credentials tab (source: personal collection)

- Step 6: Run the IAM Policy Simulator, choose the “Permissions policies” tab
- Step 7: Open a new browser tab and search for “IAM Policy Simulator”
- Step 8: On the “IAM Policy Simulator” page, choose “user-1”

- Step 9: In the “Policy Simulator” pane, select the drop-down menu, choose “Identity and Access Management”
- Step 10: on the “Select actions” drop-down menu, select the options:
 - “DeleteGroup”
 - “DeleteRolePolicy”
- Step 11: Choose “Run Simulation”
- Step 12: Close the simulator tab

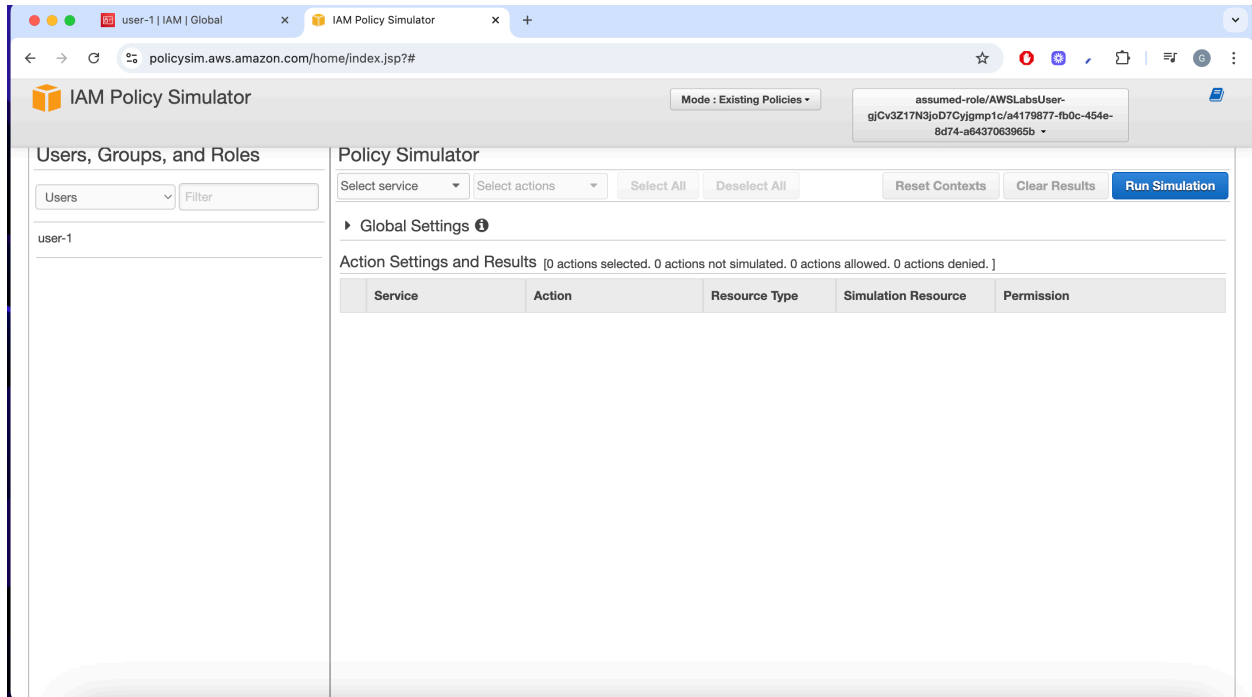


Figure 6: IAM policy simulation dashboard (source: personal collection)

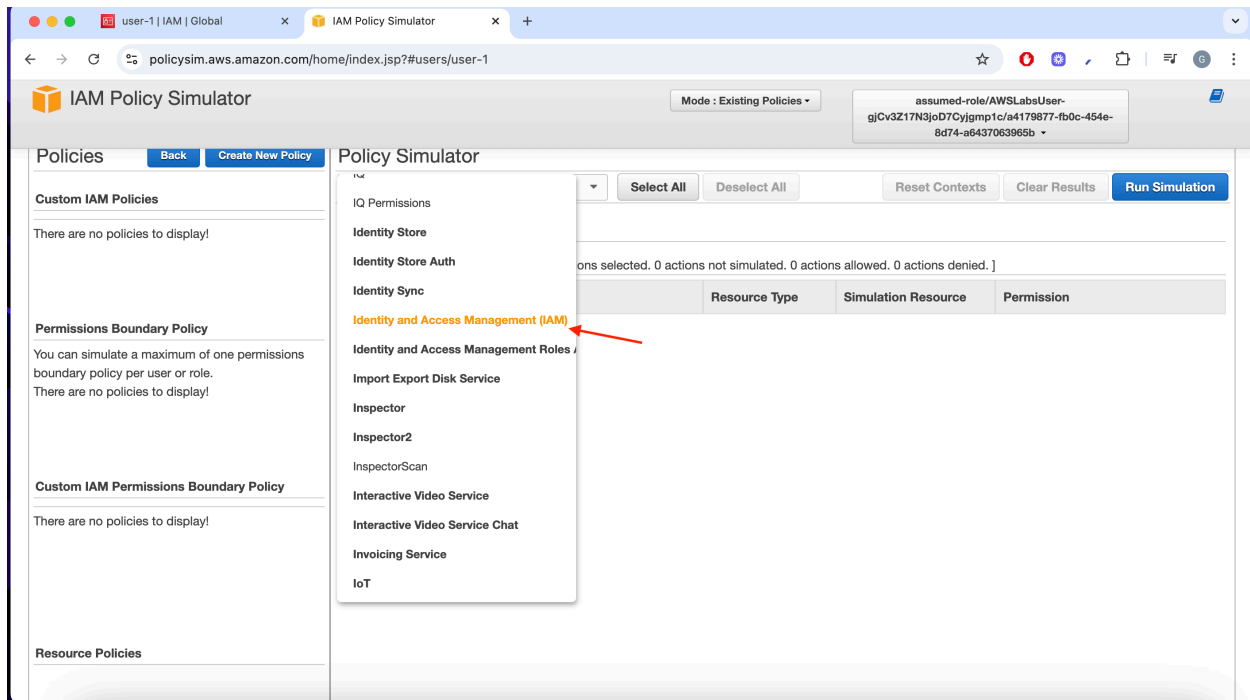


Figure 7: IAM Simulation Policy Simulator configuration (source: personal collection)

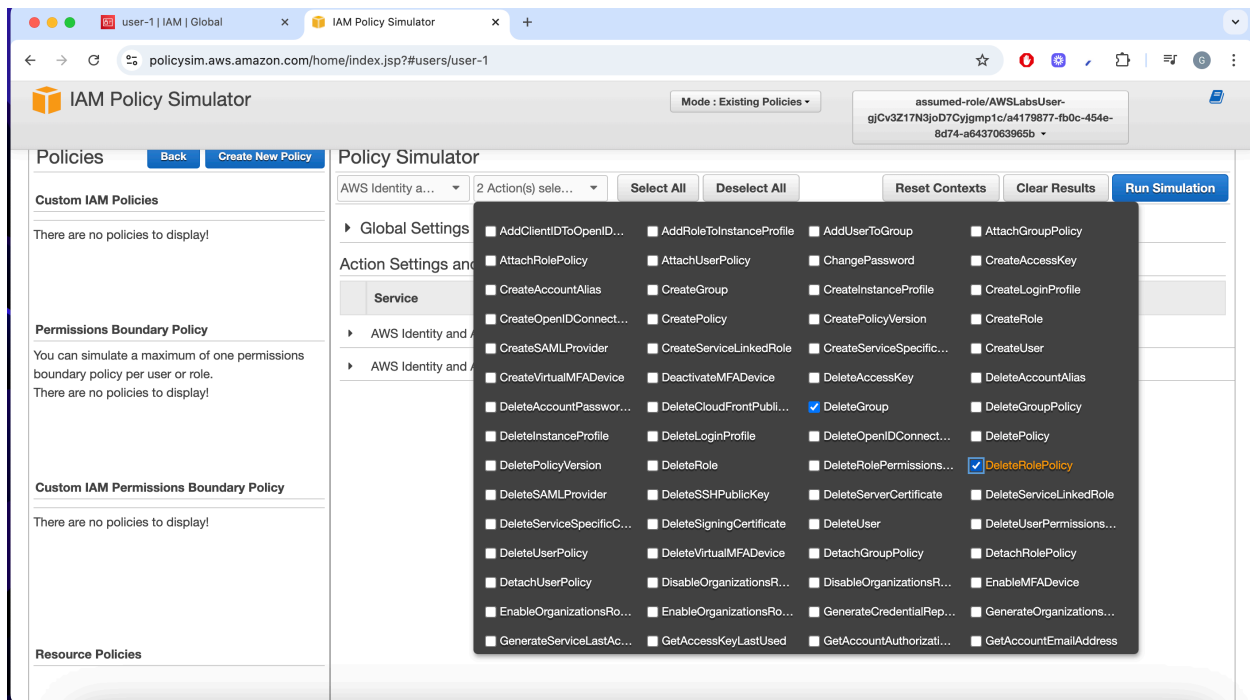


Figure 8: Select Action drop-down menu configuration (source: personal collection)

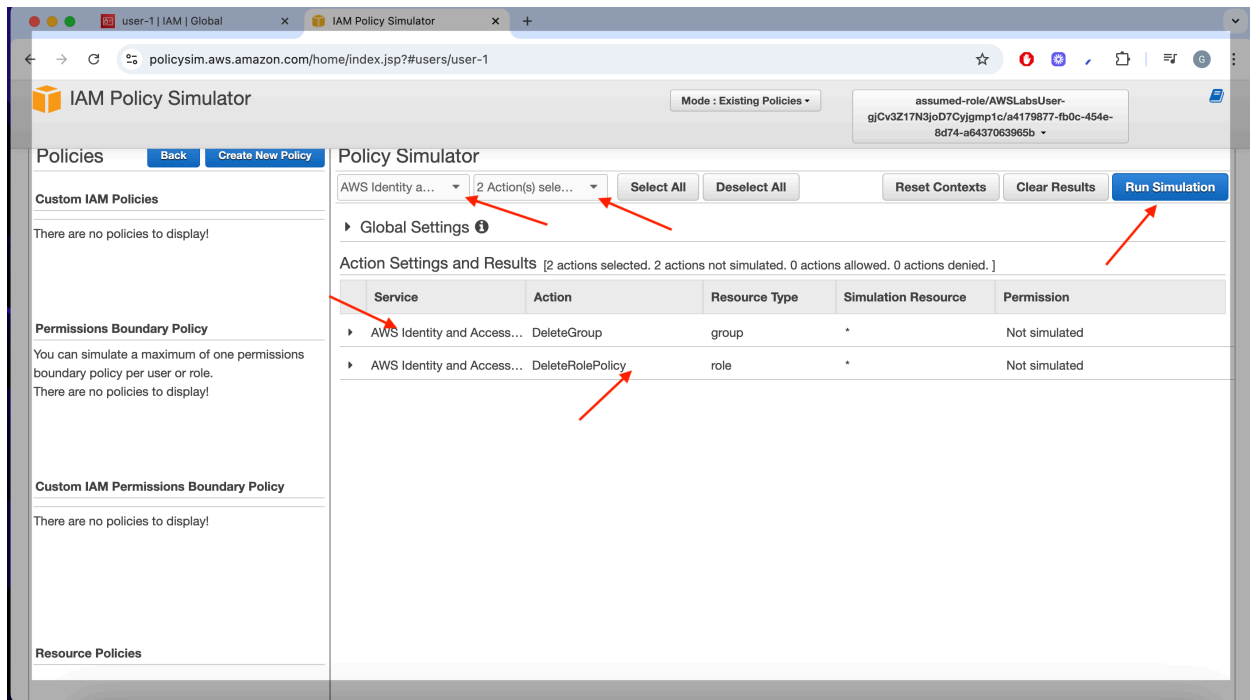


Figure 9: Action settings and Results overview (source: personal collection)

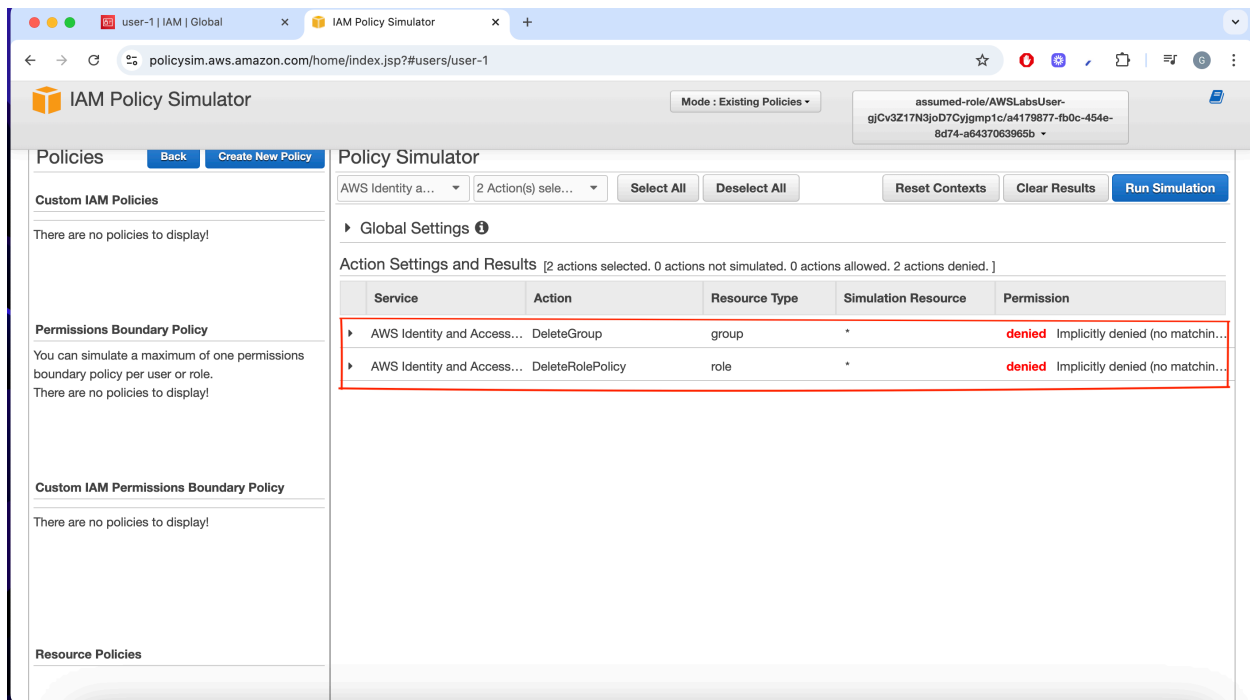


Figure 10: Access denied results (source: personal collection)

Task 2: Review the security configuration of Amazon CE2 instances

Amazon Elastic Compute Cloud (Amazon EC2) is a web service that enables you to resize compute capacity in the cloud (AWS, n.d). This is designed to make web-scale computing easier for developers.

- Step 1: Look for and select “EC2”
- Step 2: Choose “Instances” under “Instances” page
- Step 3: Review the “Network & Security” and choose “Security Groups”
- Step 4: Select the “WebServerSG” and notice the “Inbound rules”
- Step 5: Select the “BastionSG” and notice the “Inbound rules”
- Step 6: Select the “SQLSG” and notice the “Inbound rules”

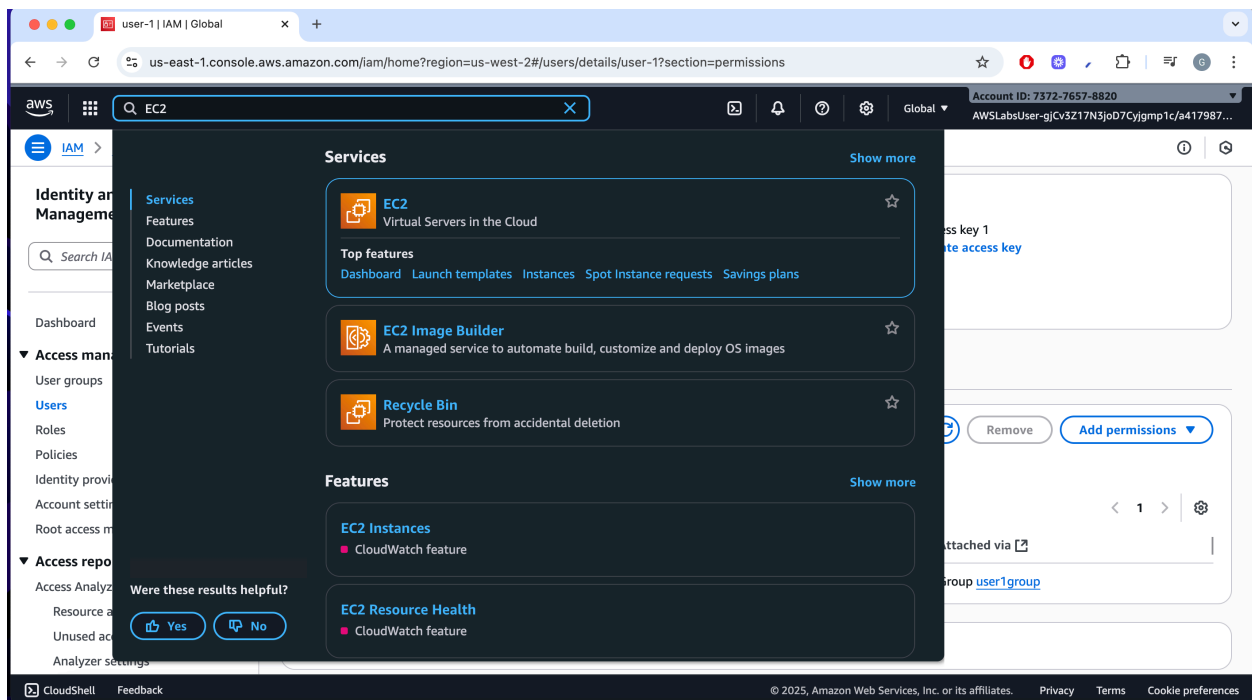


Figure 11: EC2 search (source: personal collection)

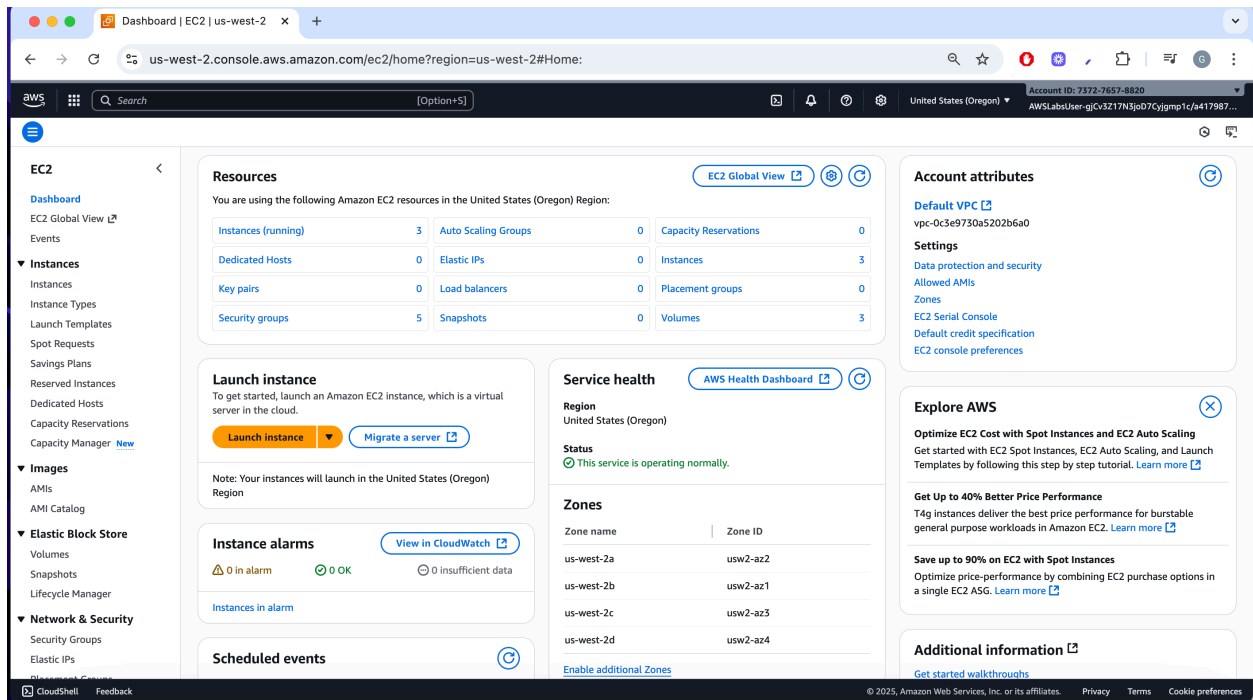


Figure 12: EC2 Dashboard (source: personal collection)

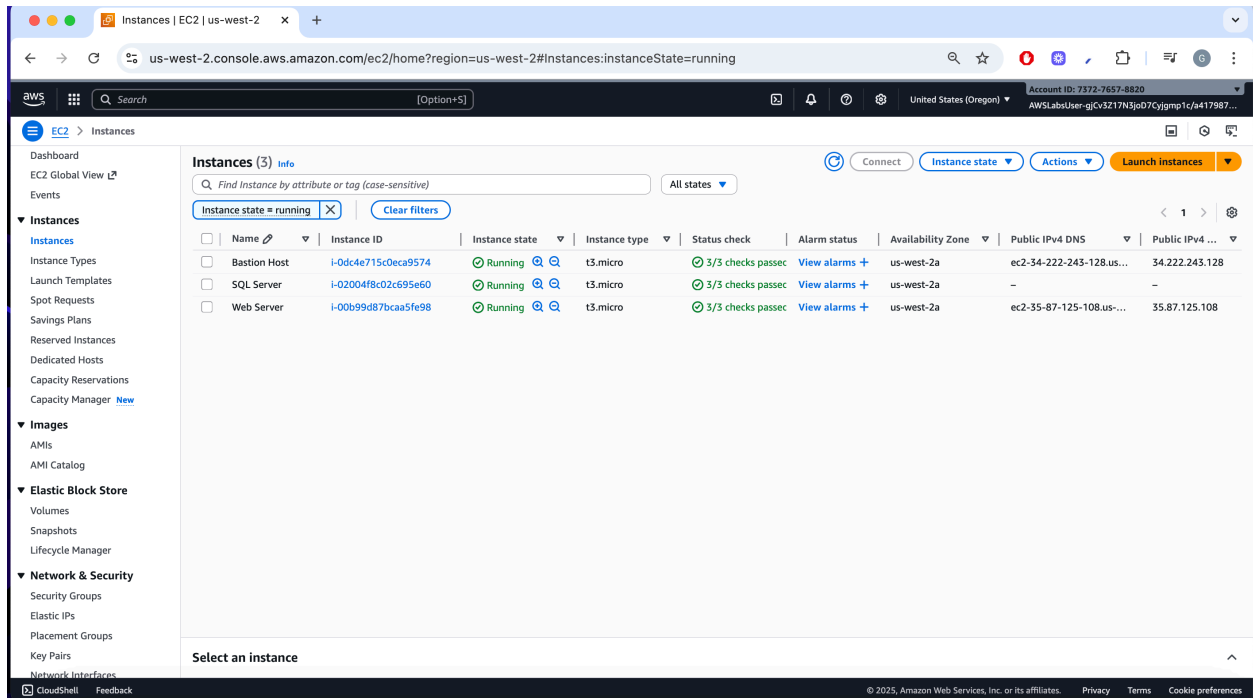


Figure 13: EC2 instances overview (source: personal collection)

Security groups | EC2 | us-west-2

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#SecurityGroups:

Security Groups (1/5)

Name	Security group ID	Security group name	VPC ID	Description	Owner
☒ WebServerSG	sg-0f03b093785cb91fb	WebSecuritySG	vpc-07efe9c5a348864de	Enable access to the Web Server	737276
☐ SQLSG	sg-068150989e776637f	SQLSG	vpc-07efe9c5a348864de	Enable access to the Bastion Server and...	737276
☐ -	sg-09fbec5f55fb31b4	default	vpc-0c3e9730a5202b6a0	default VPC security group	737276
☐ BastionSG	sg-00c9f810ddb2f5a5	BastionSG	vpc-07efe9c5a348864de	Enable access to the Bastion Server	737276
☐ -	sg-0bed2f5136efe8524	default	vpc-07efe9c5a348864de	default VPC security group	737276

sg-0f03b093785cb91fb - WebSecuritySG

Details

Security group name: WebSecuritySG

Security group ID: sg-0f03b093785cb91fb

Description: Enable access to the Web Server

VPC ID: vpc-07efe9c5a348864de

Owner: 737276578820

Inbound rules count: 3 Permission entries

Outbound rules count: 1 Permission entry

Figure 14: WebserverSG overview (source: personal collection)

Security groups | EC2 | us-west-2

us-west-2.console.aws.amazon.com/ec2/home?region=us-west-2#SecurityGroups:

Security Groups (1/5)

Name	Security group ID	Security group name	VPC ID	Description	Owner
☐ WebServerSG	sg-0f03b093785cb91fb	WebSecuritySG	vpc-07efe9c5a348864de	Enable access to the Web Server	737276
☐ SQLSG	sg-068150989e776637f	SQLSG	vpc-07efe9c5a348864de	Enable access to the Bastion Server and...	737276
☐ -	sg-09fbec5f55fb31b4	default	vpc-0c3e9730a5202b6a0	default VPC security group	737276
☒ BastionSG	sg-00c9f810ddb2f5a5	BastionSG	vpc-07efe9c5a348864de	Enable access to the Bastion Server	737276
☐ -	sg-0bed2f5136efe8524	default	vpc-07efe9c5a348864de	default VPC security group	737276

sg-00c9f810ddb2f5a5 - BastionSG

Details

Security group name: BastionSG

Security group ID: sg-00c9f810ddb2f5a5

Description: Enable access to the Bastion Server

VPC ID: vpc-07efe9c5a348864de

Owner: 737276578820

Inbound rules count: 2 Permission entries

Outbound rules count: 1 Permission entry

Figure 15: BastionSG overview (source: personal collection)

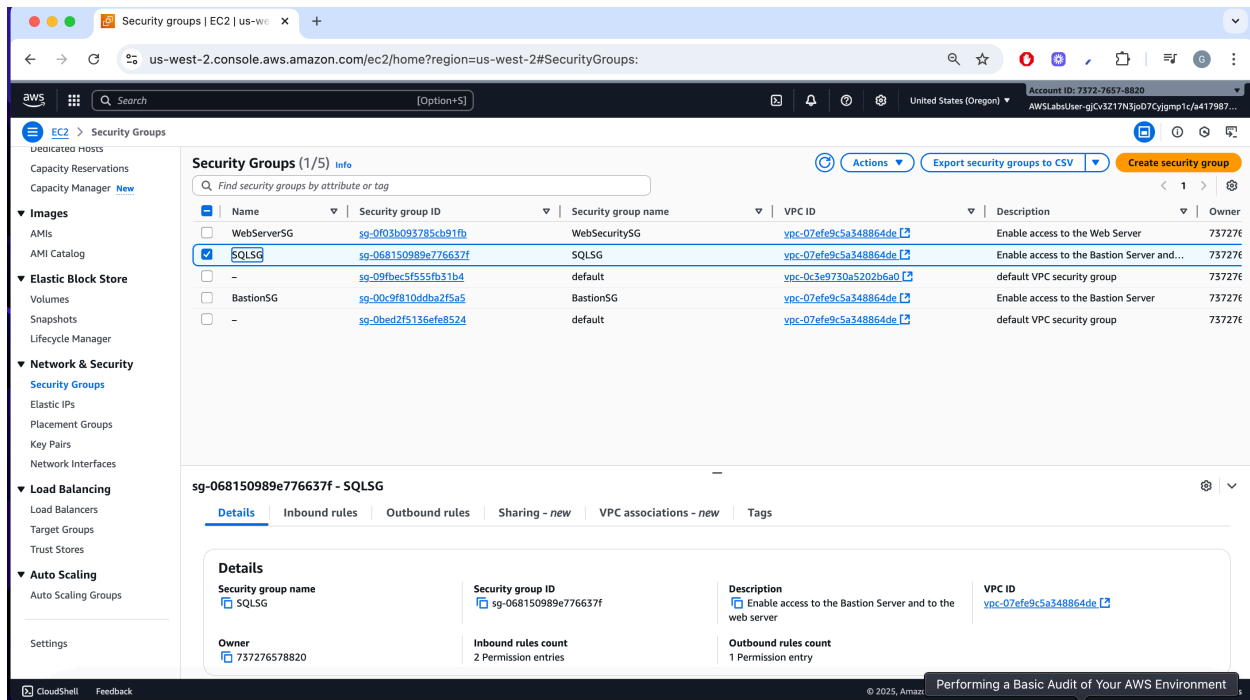


Figure 16: SQLSG overview (source: personal collection)

Task 3: Review Amazon VPC security configurations

Amazon Virtual Private Cloud (Amazon VPC) enables you to launch AWS resources into a virtual network that is defined (AWS, n.d). It includes features such as “Security Groups” and “Network Access Control Lists” (ACLs).

- Security Groups: act as a firewall for EC2 instances, controlling both inbound and outbound traffic at the instance associated to it (AWS, n.d).
 - Network Access Control Lists (ACLs): act as a firewall for associated subnets, controlling both inbound and outbound traffic at the subnet level (AWS, n.d).
-
- Step 1: Search for and choose “VPC”
 - Step 2: Choose “Your VPCs” under the “Virtual private cloud” page
 - Step 3: Select “Lab VPC”
 - Step 4: Notice the “VPC ID”, “Main Network ACL” link, and the “inbound rules” and “outbound rules”

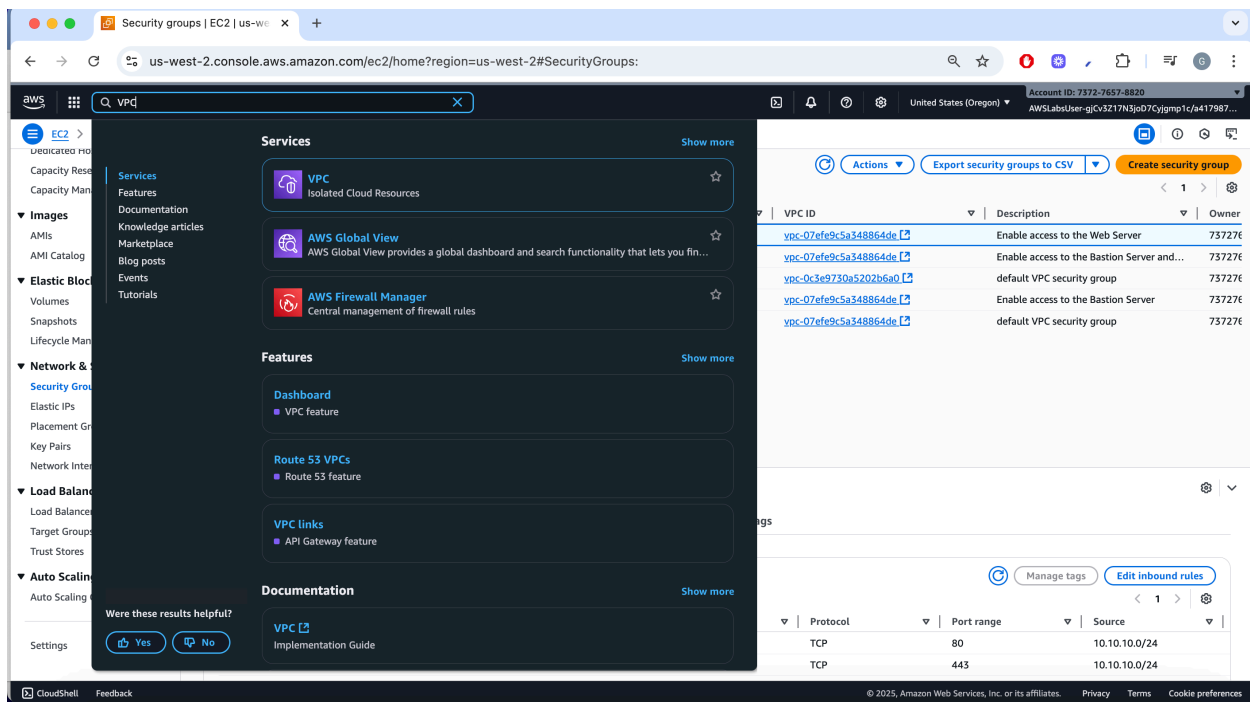


Figure 17: VPC Search (source: personal collection)

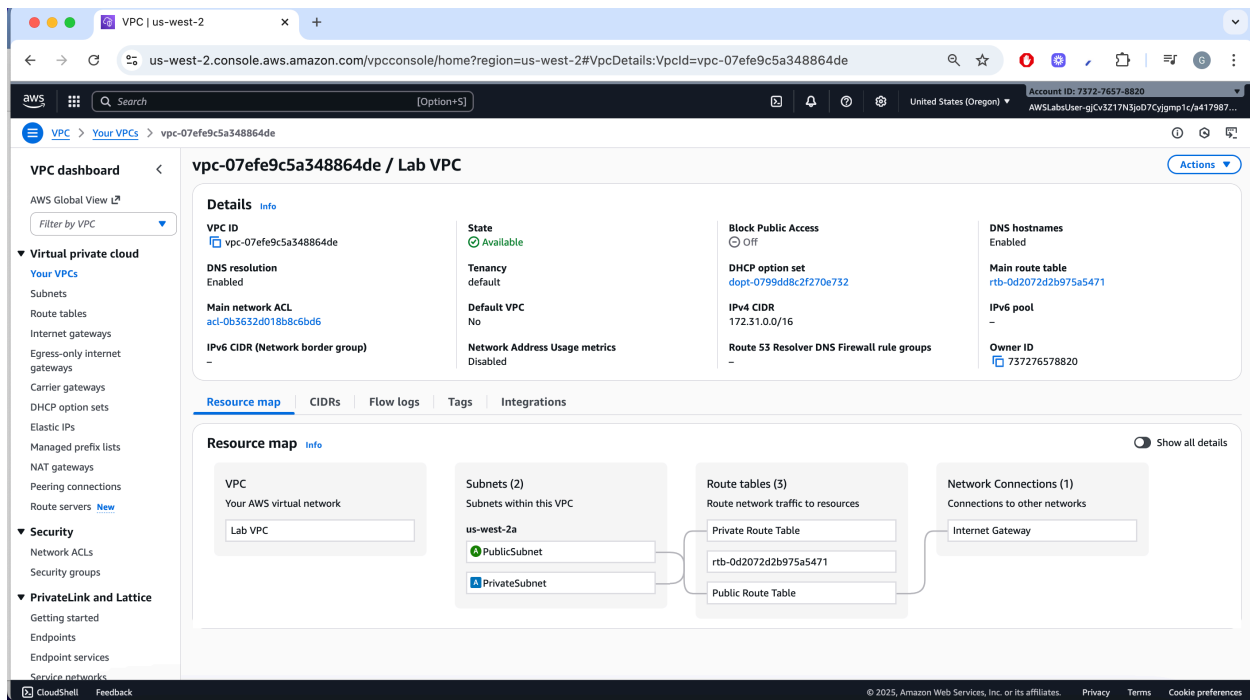


Figure 18: VPC Lab overview (source: personal collection)

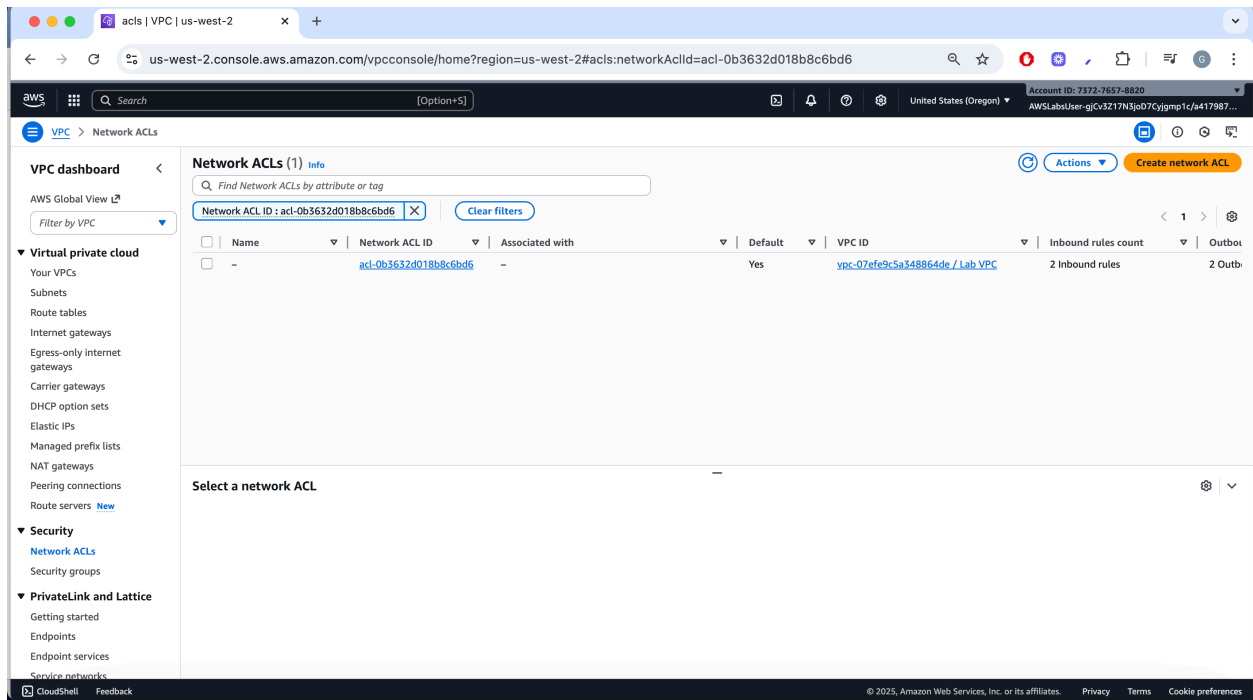


Figure 19: VPC Network ACLs overview (source: personal collection)

Task 4: Audit CloudWatch metrics and alarms

In this task, I reviewed built-in CloudWatch metrics, alarms, and service health associated with running instances, storage volumes, and data services available in the auditing instance.

Amazon CloudWatch is a monitoring and management service built for developers, system operators, site reliability engineers (SREs), and IT managers (AWS, n.d).

- Step 1: Search for and choose “CloudWatch”
- Step 2: Choose “EC2”
- Step 3: choose “per-instance Metrics”
- Step 4: in the search box, search for “CPUUtilization”
- Step 5: select “SQL Server”
- Step 6: choose “Graphed metrics” tab

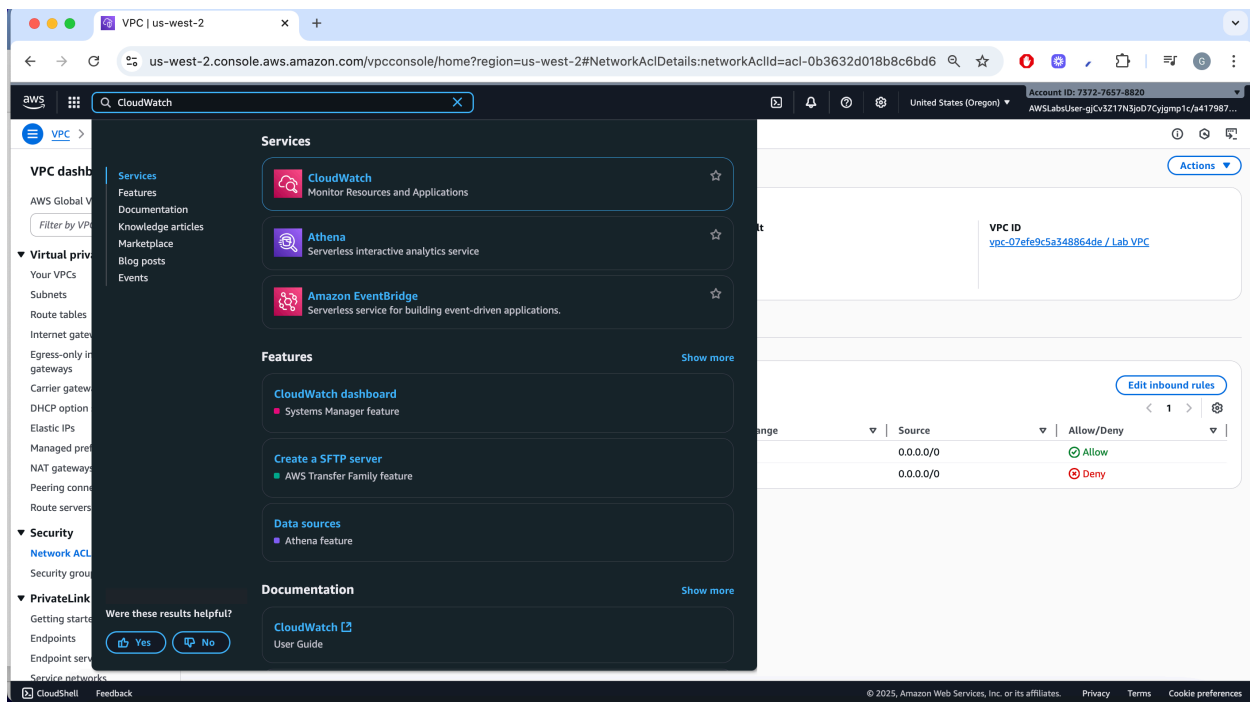


Figure 20: CloudWatch search (source: personal collection)

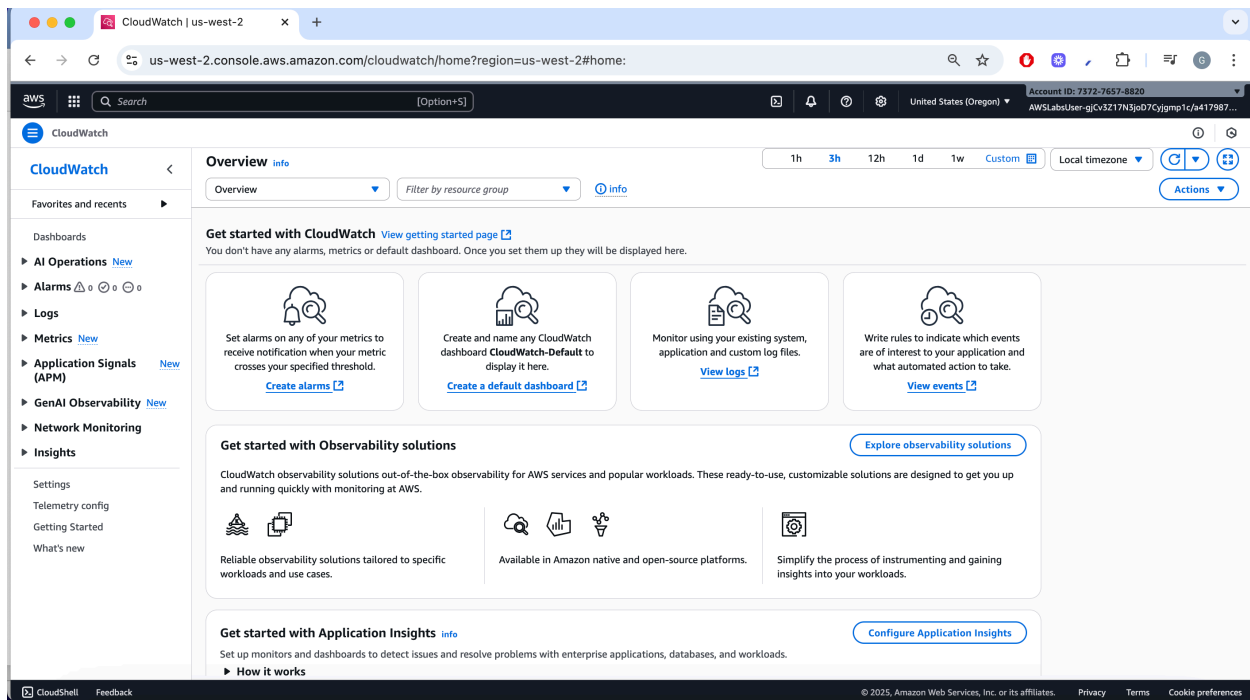


Figure 21: CloudWatch overview page (source: personal collection)

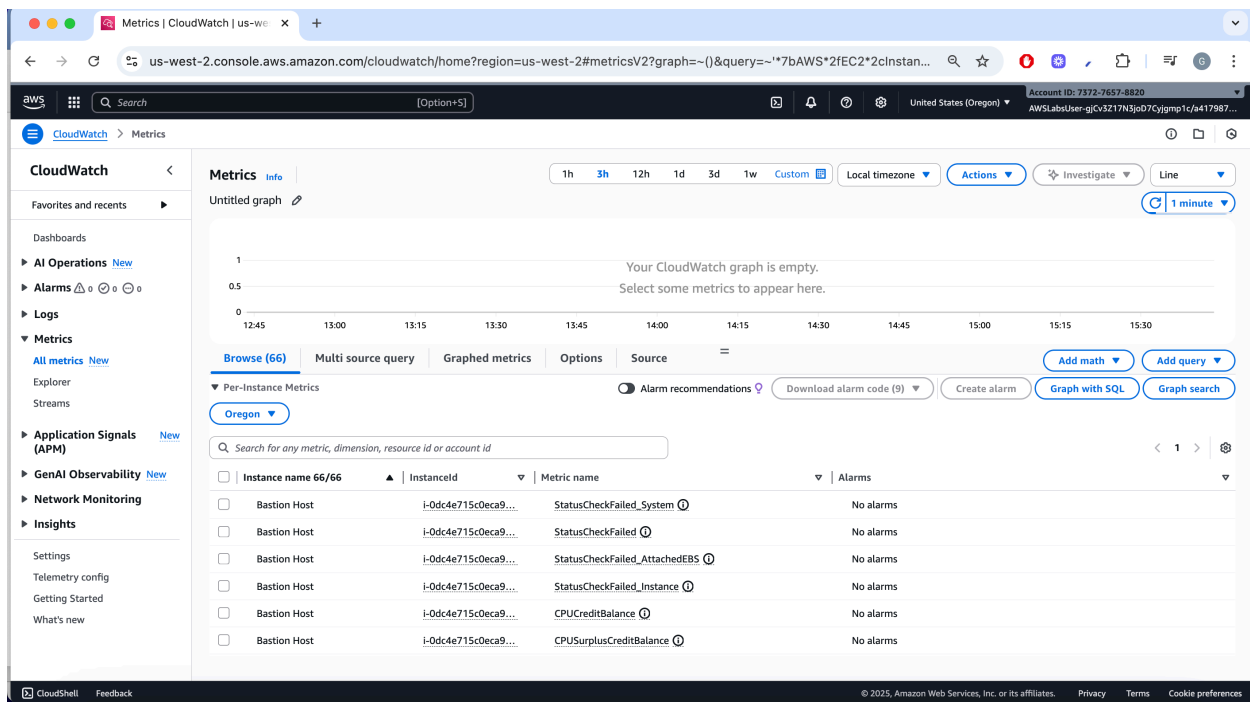


Figure 22: Per-instance Metrics (source: personal collection)

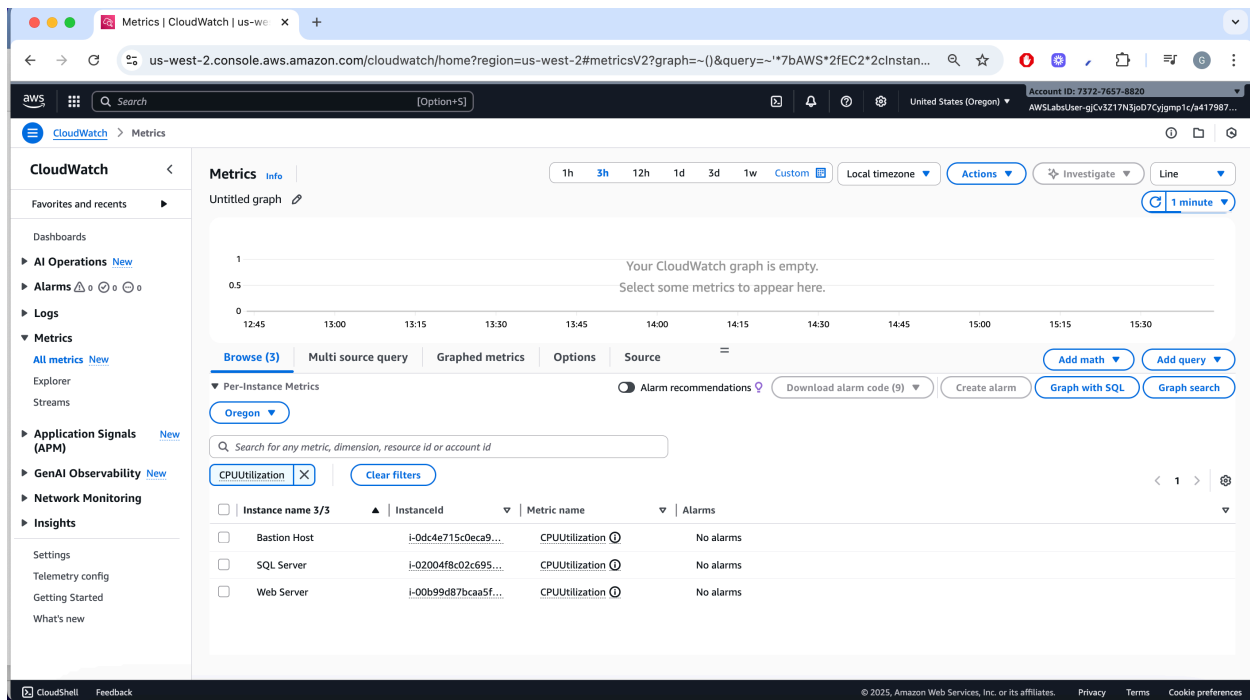


Figure 23: CPUUtilization overview (source: personal collection)

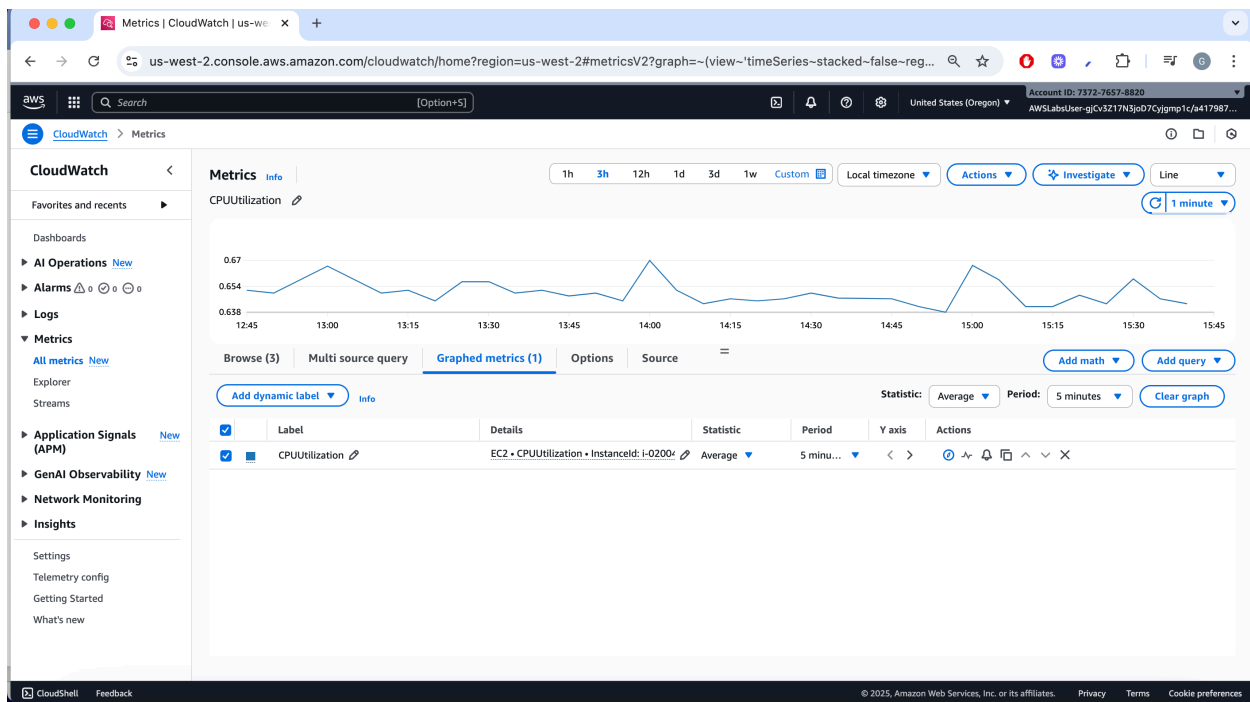


Figure 24: Graphed metrics overview (source: personal collection)

The screenshot displays the AWS Elastic Block Store (EBS) console. The left sidebar shows the navigation menu with options like Dashboard, EC2 Global View, Events, Instances, Images, Elastic Block Store, and Network & Security. The main content area shows the 'Volumes' page. A table lists three volumes with columns for Name, Volume ID, Type, Size, IOPS, Throughput, Snapshot ID, Source volume ID, Created, and Availability. The table shows three volumes of type 'gp2' with sizes of 30 GiB and IOPS of 100. Below the table, there's a section for 'Fault tolerance for all volumes in this Region' and a 'Snapshot summary' showing '0 / 3' recently backed up volumes. The top navigation bar includes the AWS logo, search bar, and account information.

Name	Volume ID	Type	Size	IOPS	Throughput	Snapshot ID	Source volume ID	Created	Availability
	vol-0b3fbf496e595f79b	gp2	30 GiB	100	-	snap-0ed4465...	-	2025/10/25 09:59 GMT+3	usw2-az2
	vol-030e23d0b543b38c8	gp2	30 GiB	100	-	snap-0ed4465...	-	2025/10/25 09:59 GMT+3	usw2-az2
	vol-011df987d823ec7eb	gp2	30 GiB	100	-	snap-0ed4465...	-	2025/10/25 09:59 GMT+3	usw2-az2

Figure 25: Elastic Block Store page (source: personal collection)

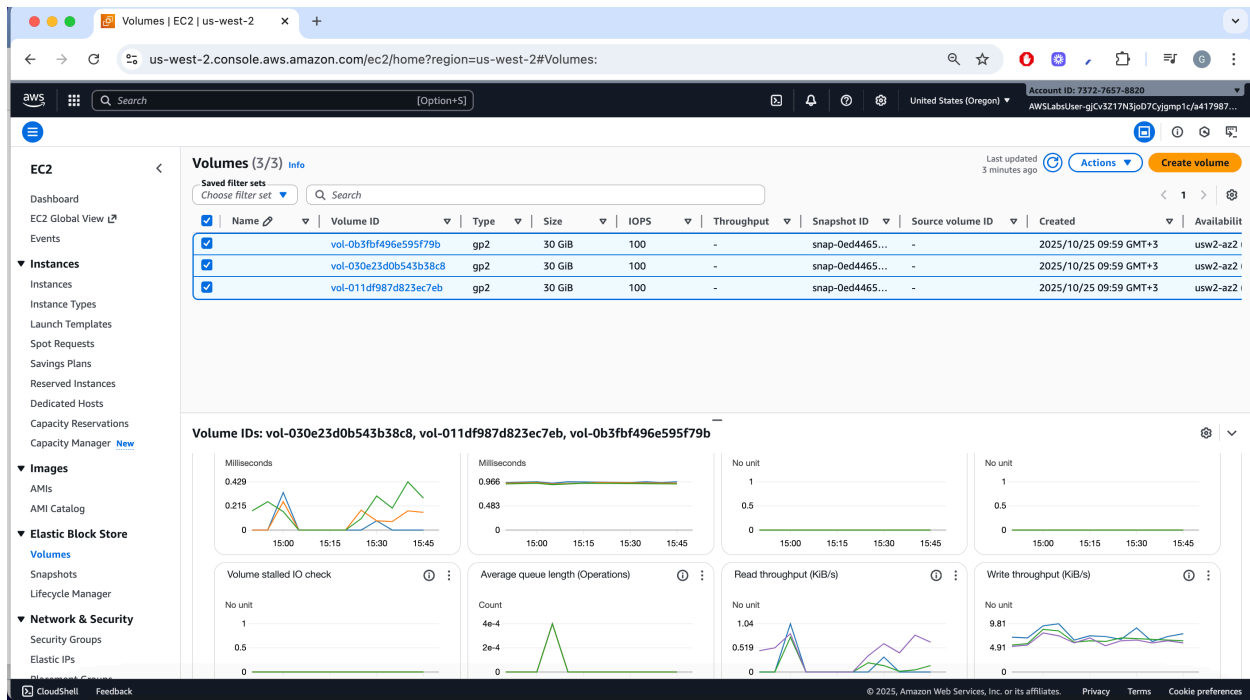


Figure 26: Volumes monitoring (source: personal collection)

Task 5: Audit CloudTrail logs

AWS CloudTrail is a service that provides governance, compliance, operational auditing, and risk auditing of your AWS account (AWS, n.d).

- Step 1: Search for and Choose “CloudTrail”
- Step 2: Choose “Trails” from the navigation pane
- Step 3: Choose the “LabCloudTrail” link to read details
- Step 4: Access the “S3 bucket”
- Step 5: Choose the link for the bucket name
- Step 6: Choose the “AWSLogs/” link
- Step 7: Choose “Open”
- Step 8: Close the lab

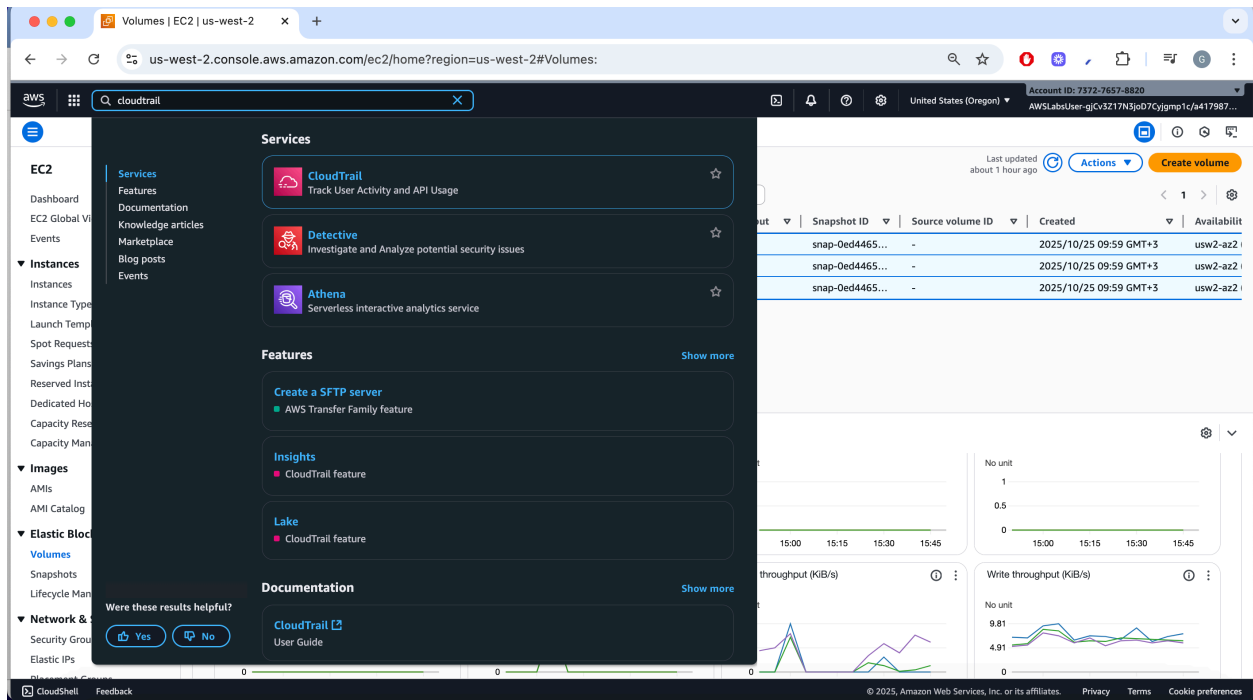


Figure 27: CloudTrail search (source: personal collection)

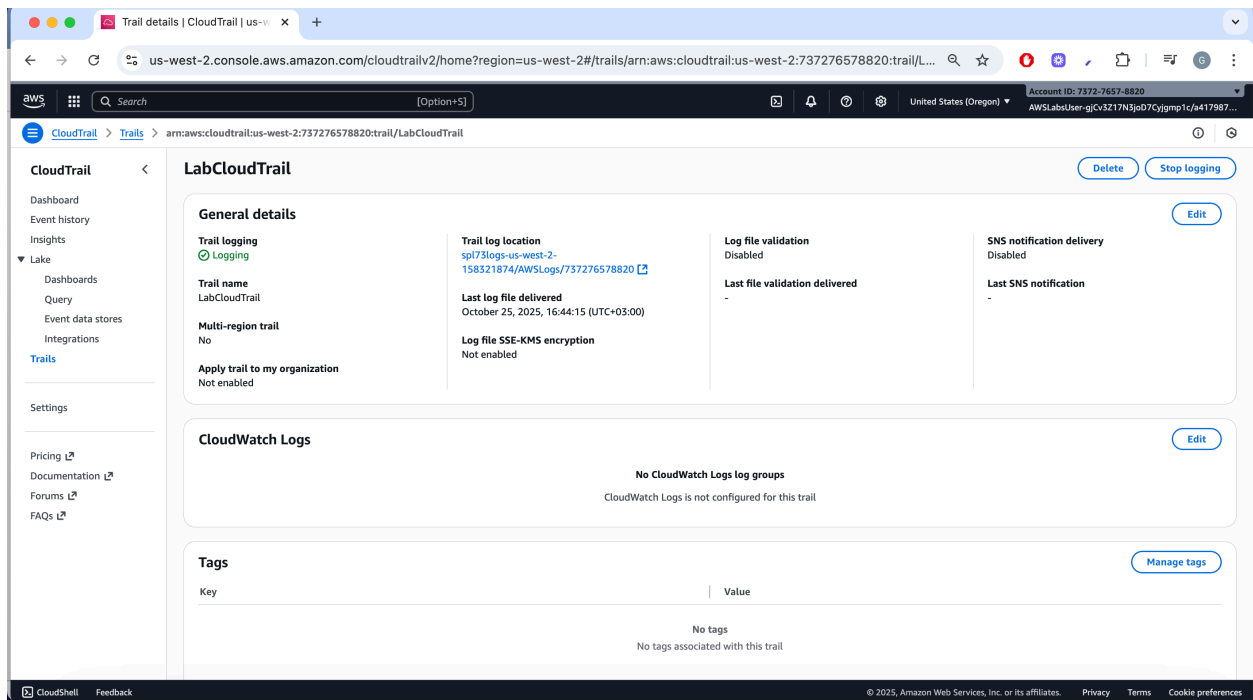


Figure 28: CloudTrail overview (source: personal collection)

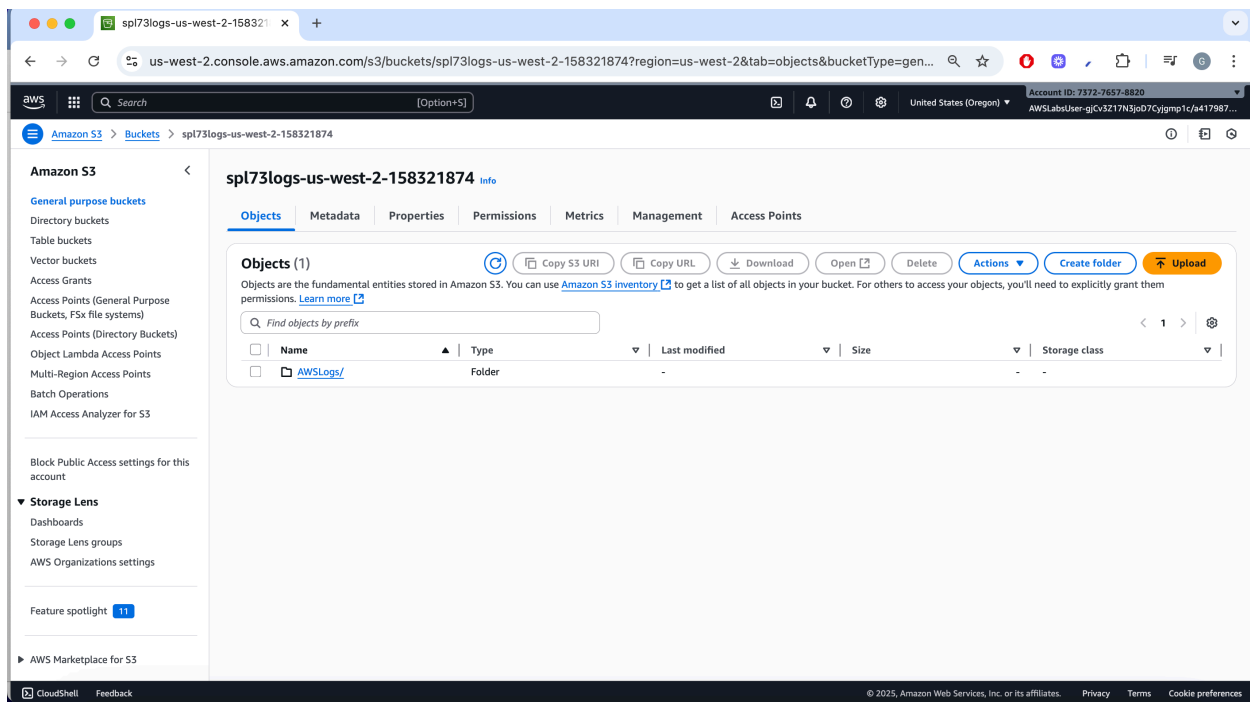


Figure 29: Buckets overview (source: personal collection)

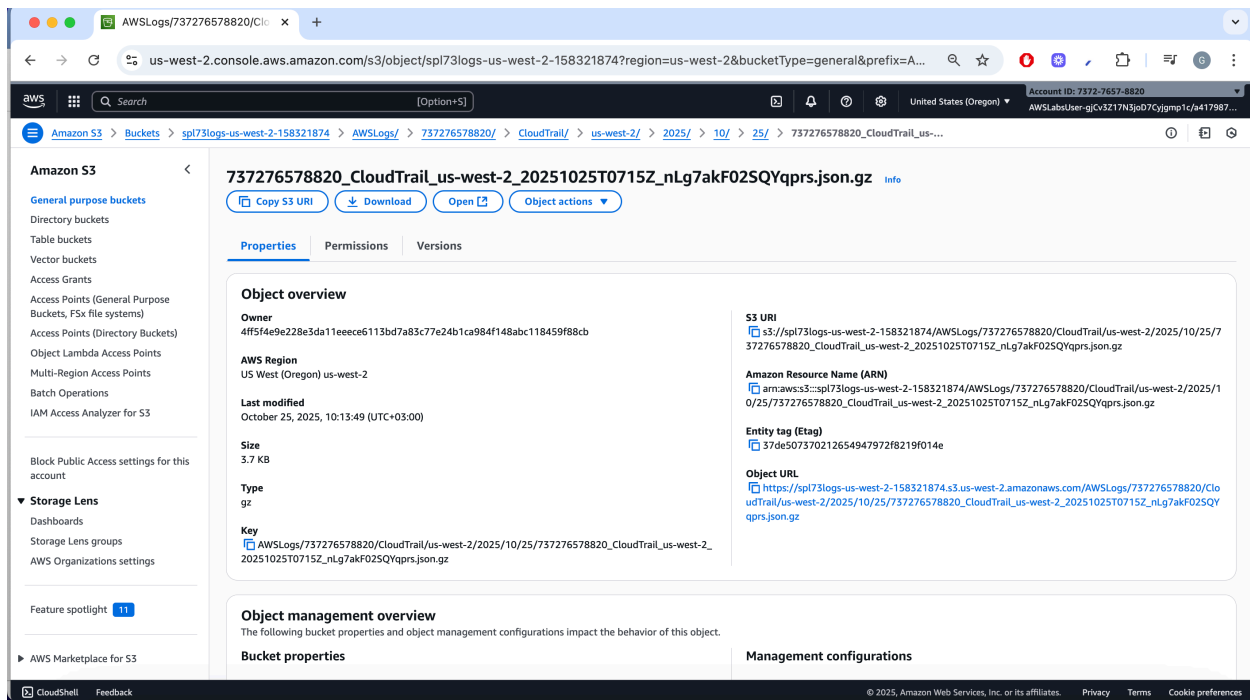


Figure 30: json.gz log (source: personal collection)

Conclusion

In this lab, I reviewed user permissions in AWS Identity and Access Management (IAM). I captured audit evidence using the AWS IAM Policy Simulator. I also examined the inbound and outbound networking rules for Amazon EC2 security groups. Additionally, I reviewed the Amazon Virtual Private Cloud (VPC) configurations, including subnets and network access control lists (ACLs). I assessed Amazon CloudWatch performance metrics and analyzed raw Amazon CloudTrail logs stored in Amazon S3.

Reference

AWS Skill Builder. *Performing a Basic Audit of your AWS Environment*. Accessed [25 Oct. 2025]. <https://lab.builder-labs.skillbuilder.aws/sa/lab/arn%3Aaws%3Alearningcontent%3Aus-east-1%3A470679935125%3Ablueprintversion%2Fspl-73%3A1.3.14-5957b1fd/en-US>.